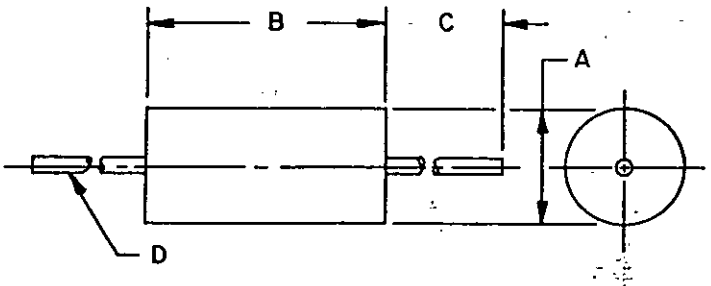
MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICRO-MINIATURE (POWDERED IRON CORE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-39010.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.146 (3.71)	.166 (4.22)
B	.365 (9.27)	.385 (9.78)
C	1.250(31.75)	1.626(41.30)
D	.023 (.58)	.027 (.69)

NOTE: Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Material: Iron core.

Weight: 0.90 gram maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202 test voltage: 1,000 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202 (70,000 feet), test voltage 200 volts rms.

FSC 5950

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Electrical characteristics: See tables I and II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part number: M39010/07 (dash number from table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A5R1**	5.1	5	45	7.9	60	0.32	495
A5R6**	5.6	5,10	45	7.9	60	0.32	495
A6R2**	6.2	5	45	7.9	55	0.50	395
A6R8**	6.8	5,10	50	7.9	55	0.50	395
A7R5**	7.5	5	50	7.9	50	0.60	360
A8R2**	8.2	5,10	50	7.9	50	0.60	360
A9R1**	9.1	5	50	7.9	45	0.90	290
A100**	10.0	5,10	55	7.9	45	0.90	290
A110**	11.0	5	55	2.5	42	1.10	265
A120**	12.0	5,10	65	2.5	42	1.10	265
A130**	13.0	5	65	2.5	40	1.40	240
A150**	15.0	5,10	65	2.5	40	1.40	240
A160**	16.0	5	65	2.5	34	2.25	185
A180**	18.0	5,10	75	2.5	34	2.25	185
A200**	20.0	5	75	2.5	30	2.50	175
A220**	22.0	5,10	75	2.5	30	2.50	175
A240**	24.0	5	60	2.5	25	2.60	170

See footnotes at end of table.

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TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency min (MHz)	DC resistance (25°C) max (ohms)	Rated dc current (mA) 2/
A270**	27.0	5,10	60	2.5	25	2.60	170
A300**	30.0	5	60	2.5	19	3.00	165
A330**	33.0	5,10	65	2.5	19	3.00	165

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e.g., M, P, R, S) and will be added to the end of the dash number
2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from initial measurement		Allowable % from specified minimum value in electrical characteristics (initial) table	
	Inductance	DC resistance	Self-resonant frequency	Q
	Percent			
Qualification inspection				
Group II	±2	---	---	-10
Group IV	±5	±(3%+.001 ohm)	1/	-10
Group V	±5	±(2%+.001 ohm)	1/	-15
Quality conformance inspection				
Group C				
Subgroup 1	±5	±(2%+.001 ohm)	1/	-15
Subgroup 3	±2	---	---	-10
Subgroup 4	±5	±(3%+.001 ohm)	1/	-10

1/ The self-resonant frequency shall be not less than the value specified in table I.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, SH
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Army - ER

(Project 5950-0503-7)

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